

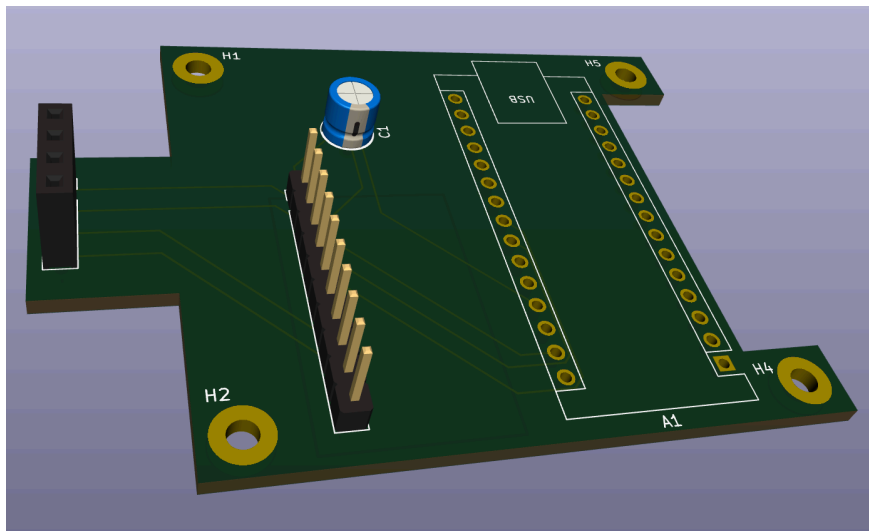
PRESSURE SENSOR BOARD v1 — DESIGN OVERVIEW

Alex Sogas

Purdue Orbital · Propulsion Electronics Subteam · Spring 2025

1. Summary

A compact 2-layer carrier board for an Arduino Nano with a plug-in connector for an I²C pressure sensor. Designed in KiCad as part of the data acquisition system for the team's solid rocket motor (Jalapeño) test program.



2. Specifications

The board measures 80 × 70 mm on two copper layers and carries an Arduino Nano v3.x running at 5V. The sensor connects through a 4-pin socket carrying power, ground, and the two I²C data lines (SDA and SCL). Five M3 mounting holes secure it to the test fixture. All design work, error checking, 3D modeling, and manufacturing file export were done in KiCad.

3. Components

- **A1 – Arduino Nano.** Reads the sensor and sends the data out over USB.
- **J1 – 4-pin socket.** The sensor plugs in here: power, ground, and two data wires.
- **C1 – 10 µF capacitor.** Smooths the 5 V supply going to the sensor.
- **J3 – 10-pin header.** Brings the sensor lines and the Arduino's I²C pins out to a header for jumpers and probing.
- **H1–H5 – Mounting holes.** Hold the board to the fixture.

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4. Connections

J1 pin 1 → 5 V and pin 2 → ground.

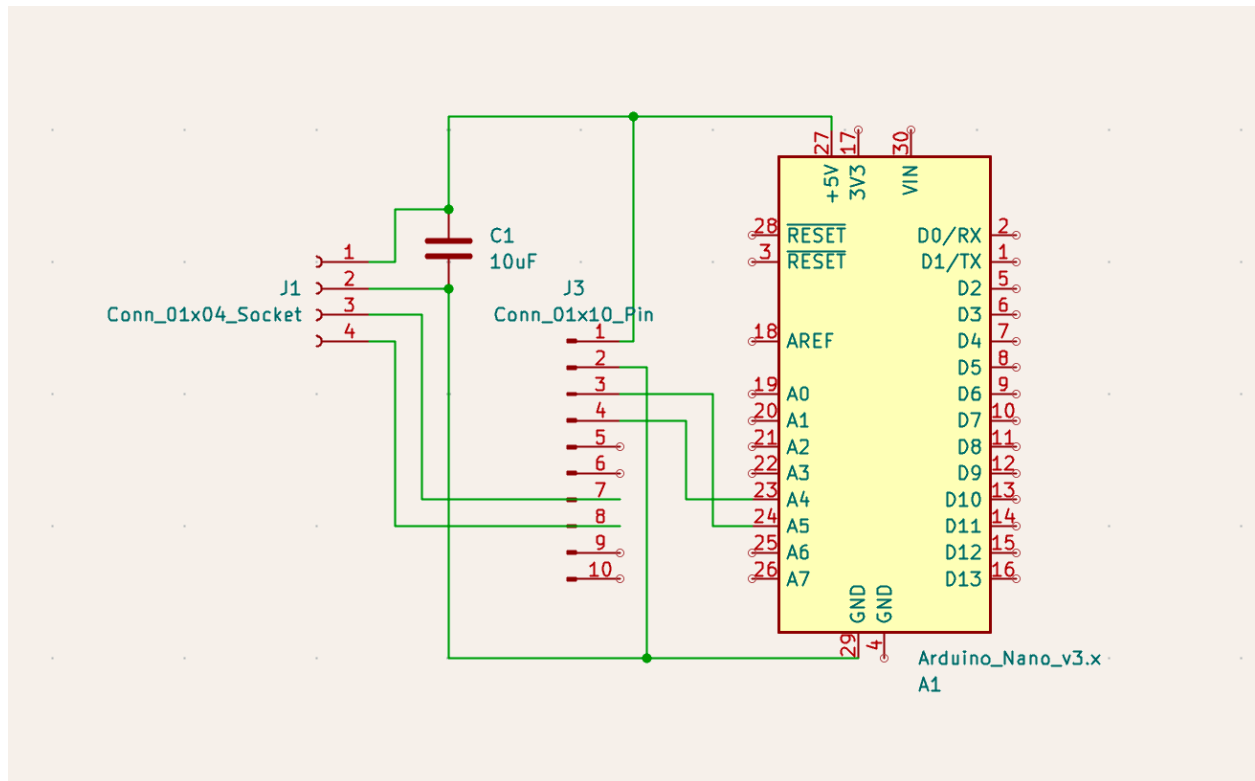
J1 pins 3 and 4 carry the sensor's data lines to J3 pins 7 and 8.

J3 pins 3 and 4 → the Arduino's A4 and A5, which are its I²C pins (SDA and SCL).

J3 pin 1 → 5 V and pin 2 → ground.

Jumpers between J3 pin 7 and 3, and pins 8 and 4 complete the path from the sensor to the Arduino.

5. Schematic

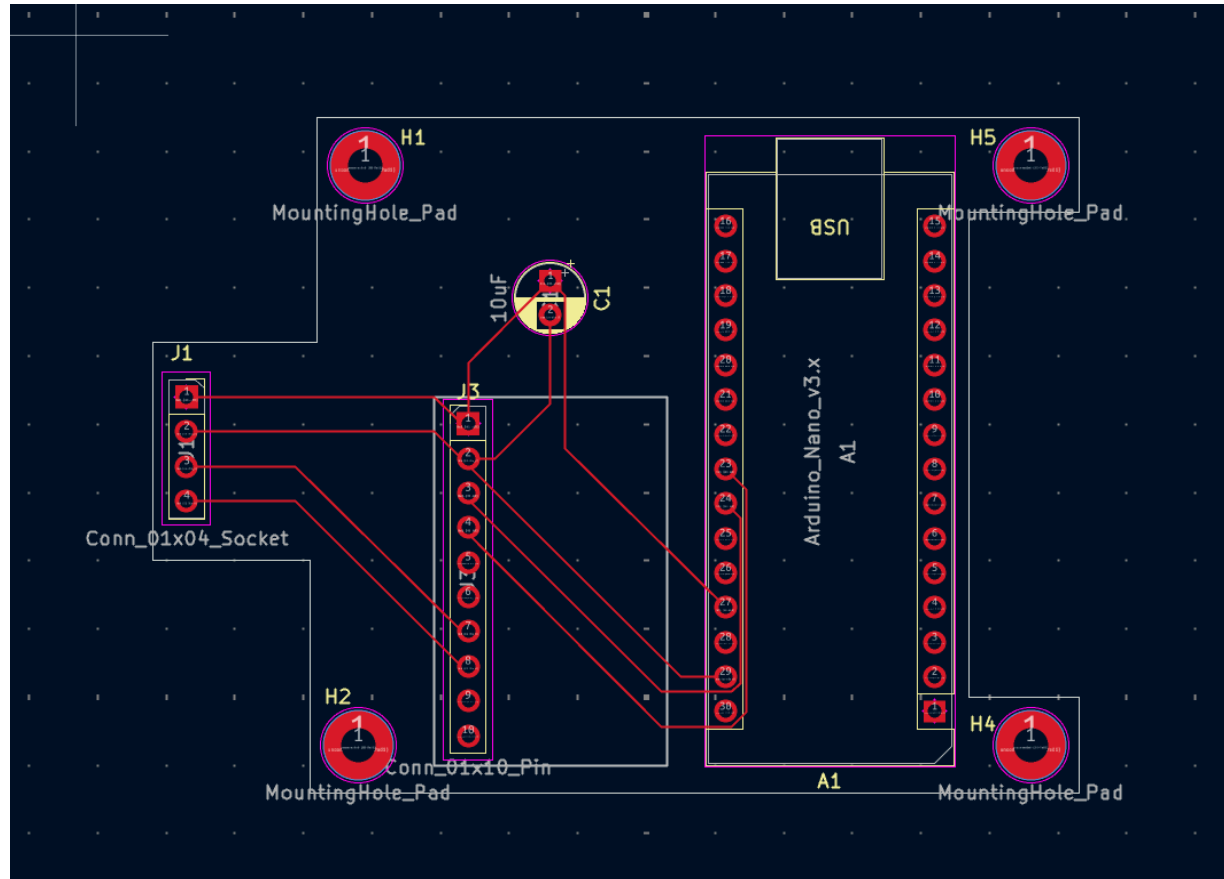


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6. Board Layout



7. Scope of Work

I translated a hand-drawn sketch from the propulsion electronics lead into a complete schematic and PCB layout, ran electrical and design rule checks, produced the 3D model, and exported manufacturing files (Gerbers) for in-house fabrication.

8. Status and Next Steps

The team etched and assembled the boards in-house after design handoff; I did not have the opportunity to test the finished board. A version 2, built around a higher-range chamber-pressure transducer, is in progress.